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Research Interests

Hardware Security, Microarchitecture Security, Side Channel Attacks

Education

Ph.D. in Control Theory and Control Engineering 2017 – 2022

Zhejiang University, USSLab

Advisor: Prof. Wenyuan Xu

B.S. in Electrical Engineering and Automation 2013 – 2017

Xi'an Jiao Tong University

Experience

Hardware Security Engineer 2022 – Present

Huawei Technologies Co., Ltd.

Publications

Journal Papers

- **[J4]** Yan Jiang, Xiaoyu Ji, **Juchuan Zhang**, Yancheng Jiang, Shui Jiang, Wenyuan Xu. “CapSpeaker: Injecting Commands to Voice Assistants Via Capacitors.” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2024.
- **[J3]** Xiaoyu Ji, **Juchuan Zhang**, Shan Zou, Yi-Chao Chen, Gang Qu, Wenyuan Xu. “MagView++: Data Exfiltration via CPU Magnetic Signals Under Video Decoding.” *IEEE Transactions on Mobile Computing (TMC)*, 2024.
- **[J2]** Xiaoyu Ji, Yushi Cheng, **Juchuan Zhang**, Yuehan Chi, Wenyuan Xu, Yi-Chao Chen. “Device Fingerprinting with Magnetic Induction Signals Radiated by CPU Modules.” *ACM Transactions on Sensor Networks (TOSN)*, 2022.
- **[J1]** Xiaoyu Ji, Chaohao Li, Xinyan Zhou, **Juchuan Zhang**, Yanmiao Zhang, Wenyuan Xu. “Authenticating Smart Home Devices via Home Limited Channels.” *ACM Transactions on Internet of Things (TIOT)*, 2020.

Conference Papers

- [C7] Shan Zou, **Juchuan Zhang**, Shui Jiang, Yushi Cheng, Xiaoyu Ji, Wenyan Xu. “OutletGuarder: Detecting DarkSide Ransomware by Power Factor Correction Signals in an Electrical Outlet.” *ICPADS*, 2022.
- [C6] Xiaoyu Ji, **Juchuan Zhang**, Shui Jiang, Jishen Li, Wenyan Xu. “CapSpeaker: Injecting Voices to Microphones via Capacitors.” *ACM CCS*, 2021.
- [C5] **Juchuan Zhang**, Xiaoyu Ji, Yuehan Chi, Yi-Chao Chen, Bin Wang, Wenyan Xu. “OutletSpy: Cross-Outlet Application Inference via Power Factor Correction Signal.” *ACM WiSec*, 2021.
- [C4] **Juchuan Zhang**, Xiaoyu Ji, Wenyan Xu, Yi-Chao Chen, Yuting Tang, Gang Qu. “MagView: A Distributed Magnetic Covert Channel via Video Encoding and Decoding.” *IEEE INFOCOM*, 2020.
- [C3] Yushi Cheng, Xiaoyu Ji, **Juchuan Zhang**, Wenyan Xu, Yi-Chao Chen. “DeMiCPU: Device Fingerprinting with Magnetic Signals Radiated by CPU.” *ACM CCS*, 2019.
- [C2] Chaohao Li, Xiaoyu Ji, Xinyan Zhou, **Juchuan Zhang**, Jing Tian, Yanmiao Zhang, Wenyan Xu. “HlcAuth: Key-free and Secure Communications via Home-Limited Channel.” *ACM AsiaCCS*, 2018.
- [C1] Zhou Zhuang, Xiaoyu Ji, Taimin Zhang, **Juchuan Zhang**, Wenyan Xu, Zhenhua Li, Yunhao Liu. “FBSleuth: Fake Base Station Forensics via Radio Frequency Fingerprinting.” *ACM AsiaCCS*, 2018.